Velocity, density, current, power and temperature gradient in physics; rate of reaction and compressibility in chemistry, rate of growth and blood velocity in biology; marginal cost and marginal profit in economics; rate of heat flow in geology; rate of improvement of performance in psychology – these are all cases of a single mathematical concept, the derivative.

This is an illustration of the fact that part of the power of mathematics lies in its abstractness. A single abstract mathematical concept (such as derivatives) can have different interpretations in each of the sciences. When we develop the properties of the mathematical concept, we can then apply these results to all of the sciences. This is much more efficient than developing properties of special concepts in each separate science.

One of the greatest creations of the ancient past was Euclidean geometry. This monumental work was not matched in importance until the discovery of calculus almost two thousand years later.

Calculus was created independently in England by **Sir Isaac Newton** (1642 – 1727) and in Germany by **Gottfried Wilhelm Leibnitz** (1646 – 1716) in the last quarter of the seventeenth century.

Newton's interest in mathematics began with his study of two of the great books on mathematics at that time: **Euclid's Elements and Descartio La Geometric**. He also became aware of the work of the great scientists who preceded him, including Galileo and Fermat.

By the end of 1664, Newton seemed to have mastered all the mathematical knowledge of the time and had begun adding substantially to it. In 1665, he began his study of the rates of change or flexions, of quantities, such as distances or temperatures that varied continuously. The result of this study was what we today call differential calculus. All who study mathematics today stand on Isaac Newton's shoulders.

Many of Leibnitz's mathematical papers appeared in the journal 'Acta Eruditorum' which he cofounded in 1682. This journal contained his work on calculus and led to the bitter controversy with Newton over who first discovered calculus. Leibnitz was the first to publish the important results on calculus and was the first to use the notation that has now become standard.



Augustin – Louis Cauchy
(1789 – 1857)

Augustin-Louis Cauchy (1789 – 1857), born in Paris in 1789 is considered to be the most outstanding mathematical analyst of the first half of the nineteenth century. Cauchy made many contributions to calculus. In his 1829 text book '**Lecons le calcul differential**', he gave the first reasonably clear definition of a limit and was the first to define the derivative as the limit of the difference quotient,

$$\frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x} .$$

Karl Weierstrass (1815-1897), a German mathematician gave the precise definition $(\in - \delta$ definition) of the concepts of limit, continuity and differentiability.

What is Calculus?

Calculus is the mathematics of ratio of change of quantities. It is also the mathematics of tangent lines, slopes, areas, volumes, arc lengths, centroids, curvatures and a variety of other concepts that have enabled scientists, engineers and economists to model real-life situations.

Although pre calculus mathematics deals with velocities, accelerations, tangent lines, slopes and so on, there is a fundamental difference between pre calculus mathematics and calculus. Pre calculus mathematics is more static, whereas calculus is more dynamic. Here are some examples:

- ★ An object travelling at a constant velocity can be analyzed with pre calculus mathematics. To analyse the velocity of an accelerating object, you need calculus.
- ★ The slope of a line can be analysed with pre calculus mathematics. To analyse the slope of a curve, you need calculus.

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